

Causal Inference With Spatial Data

Empirical Methods — Week 16

Teaching deck draft

Week 16 question

- How do spatial spillovers, sorting, exposure, and correlated shocks change causal designs?
- Lecture 15 built spatial objects; Lecture 16 asks when those objects support causal claims.
- The core discipline: name the geography, the counterfactual, the interference structure, and the inference problem.
- A map is not a design until the comparison and assumptions are visible.

Geography has three jobs

Role	What geography does	Paper template
Source	border, distance, or exposure creates variation	Black; Dell; Dube–Lester–Reich
Threat	nearby units share residual shocks	Conley
Mechanism	access, neighborhood, spillover, or local labor market is the channel	Miguel–Kremer; Chetty et al.

Good spatial papers say which job geography is doing before estimating.

Spatial clustering and Conley-style inference

$$\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1} \left[\sum_i \sum_j K\left(\frac{d_{ij}}{c}\right) x_i x_j' \hat{u}_i \hat{u}_j \right] (X'X)^{-1}$$

- Nearby residuals may share weather, labor demand, policy, infrastructure, or demographics.
- Choose coordinates, distance metric, cutoff c , and kernel transparently.
- Report sensitivity to plausible spatial cutoffs.
- This fixes a variance problem, not bias from spillovers or sorting.

Spillovers and exposure mappings

$$Y_i = Y_i(D_i, E_i) \quad E_i = \sum_{j \neq i} w_{ij} D_j$$

- Spatial work often violates simple no-interference logic.
- w_{ij} can encode distance, adjacency, commuting, school links, or market exposure.
- Miguel–Kremer is the canonical lesson: externalities are a causal object, not just noise.
- The mapping defines the estimand; it still needs a credible assignment or exposure design.

Border designs and spatial RD

$$Y_i = \alpha + \tau \mathbf{1}\{b_i \geq 0\} + f_-(b_i) \mathbf{1}\{b_i < 0\} + f_+(b_i) \mathbf{1}\{b_i \geq 0\} + X_i' \gamma + \varepsilon_i$$

- b_i : signed distance to the policy or institutional boundary.
- Black: school-boundary comparisons and local amenities.
- Dell: historical boundary and continuity of predetermined geography.
- Dube–Lester–Reich: contiguous counties and local labor-market comparisons.
- Main caveat: borders bundle institutions, sorting, spillovers, and local shocks.

Shift-share and place-based designs

$$Z_{\ell} = \sum_s w_{\ell s} g_s$$

- $w_{\ell s}$: baseline local shares; g_s : common sector, origin, product, or task shock.
- The geography ℓ defines the local exposure object.
- Diagnostics: dominant shares, effective shock variation, baseline year, leave-one-out shocks.
- Place-based policies require both treatment geography and exposure geography.
- Main caveat: polished exposure is not exogeneity.

Sorting, MAUP, and correlated shocks

- When place is the treatment, selection into place is usually the central threat.
- Chetty–Hendren–Katz show why random assignment, move timing, and exposure duration matter.
- MAUP changes the estimand when tract, county, and commuting-zone estimates differ.
- Correlated local shocks can contaminate nearby controls or create spatial residual patterns.
- Geography sensitivity should be interpreted through the mechanism, not as a ritual.

Design diagnostics checklist

- ① State the mechanism before choosing the geography.
- ② Define observation, treatment, exposure, and identifying variation separately.
- ③ Write down the interference assumption.
- ④ Check border balance, exposure support, or share decomposition.
- ⑤ Diagnose contaminated controls and correlated local shocks.
- ⑥ Report spatial inference as inference, not identification.
- ⑦ Explain the counterfactual: nearby places, untreated time, other boundaries, or lower exposure.

Reproduce

- synthetic border counties
- pair fixed effects
- treatment estimate
- robust vs Conley SE

Diagnose

- balance near borders
- cutoff sensitivity
- neighbor exposure
- role of geography

Transfer

- shift-share exposure
- dominant shares
- leave-one-out shocks
- design memo

What not to confuse

- Spatially robust standard errors are not a substitute for a spillover model.
- A border map is not proof of local continuity.
- A narrow bandwidth is not proof that sorting disappears.
- A shift-share index is not automatically an instrument.
- A stable estimate across geographies is reassuring only if each geography is meaningful.

- Spatial causal inference is ordinary identification logic under harder geography.
- Place definitions, interference structure, and estimand choice are part of identification.
- Conley-style inference solves a dependence problem in uncertainty, not a design problem in bias.
- Border, exposure, shift-share, and mover designs each require paper-specific diagnostics.
- Lecture 17 asks when spatial equilibrium makes partial-equilibrium treatment effects incomplete.